EE1060: Differential Equations and Transform Techniques

Solutions to Practice Problem Set - 05

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1. $y\left(x\right) = c_1 y_1\left(x\right) + c_2 y_2\left(x\right)$ where $y_1(x) = \frac{1}{2}\left(3x^2 - 1\right)$ and $y_2(x) = \frac{1}{2}\ln\left(\frac{1+x}{1-x}\right)\left(3x^2-1\right)$.
2. $y(x) = c_1 x^2 + \frac{c_2}{x^2}$
3. $y(x) = c_1 \frac{\sin x}{\sqrt{x}} + c_2\left(-\frac{\cos x}{\sqrt{x}}\right)$
4. $y(x) = \left(c_1 + c_2 x\right)e^{2x} + \frac{x}{2}e^{-2x}$
5. $y(x) = \left(c_1 + c_2 x\right)e^{2x} + \frac{1}{4}x^2 + \frac{1}{2}x + \frac{3}{8}$
6. $y(x) = \left(c_1 + c_2 x\right)e^{2x} + \frac{e^{2x}}{40}\left(\cos 4x - 2\sin 4x\right)$
7. $y(x) = c_1 \cos kx + c_2 \sin kx - \frac{1}{2k}\left(x\cos kx\right)$
8. $y(x) = c_1 e^{\left(-2+\sqrt{3}\right)x} + c_2 e^{\left(-2-\sqrt{3}\right)x} - \frac{1}{4k^2}\cos kx$
9. $y(x) = c_1\left(2x\right) + c_2\left(e^{-x^2}\right)$
10. $y(x) = c_1 e^{x} + c_2 e^{-x} + \left(\frac{x^2}{4} - \frac{x}{2}\right)e^{x}$
11. $\lambda = 0;\; y(x) = x$ and $\lambda = \left(\frac{n\pi}{L}\right)^2;\; y(x) = c\cos\left(\frac{n\pi}{L}x\right)$ for $n = 1, 2, \cdots$
12. $\lambda = n^2;\; y(x) = c\cos\left(nx\right)$ for $n = 1, 2, \cdots$
13. $\lambda = 1 + \left(n\pi\right)^2;\; y(x) = ce^{-x}\sin\left(n\pi x\right)$ for $n = 1, 2, \cdots$
14. $c_1 + c_2 e^{3x} + c_2 x e^{3x} + \frac{1}{6}x^2 + \frac{2}{9}x + c$